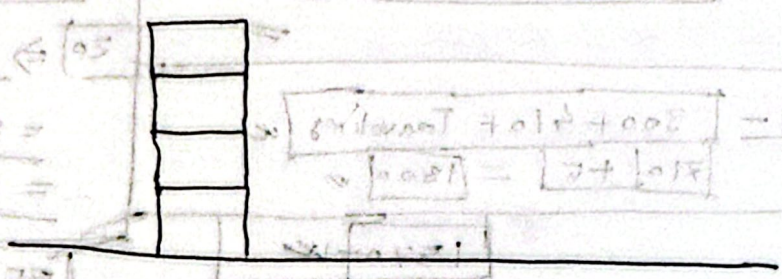
Stack

stores a List of items in which at items can be added ~~on~~ to or removed from the one end

LIFO = Last in First out — Like an Elevator.

Stack — last in First out



Piles of boxes

push (z) - Add element in the last.

pop() - remove the top element

top() - Give you the top element

empty() - will tell the Stack is empty or not

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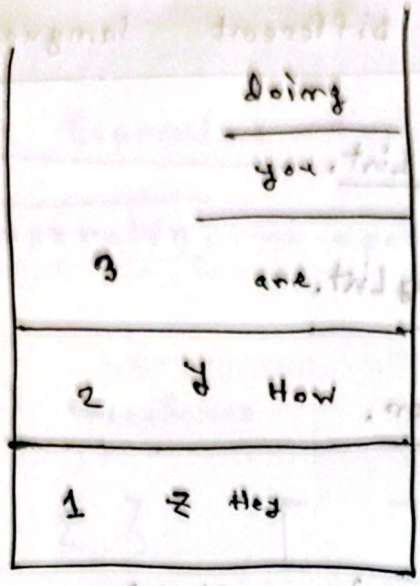
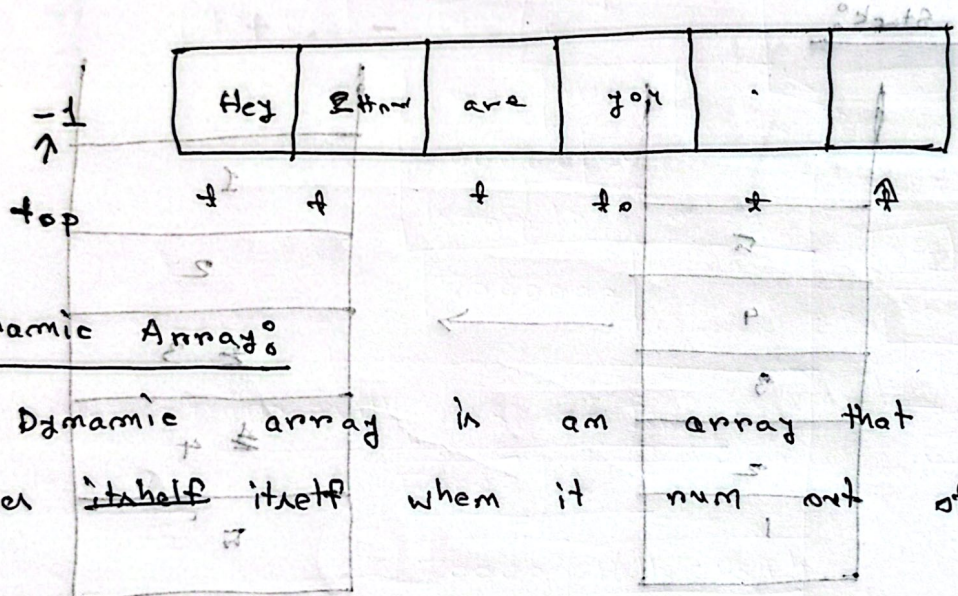


Time complexity O(1)

Array Implementation:



Dynamic Array:

Dynamic array is an array that automatically resizes ~~itshelf~~ itself when it run out of Space.

~~dynamik a~~

dynamic array grows or shrinks as you add or remove elements

Steps when array is ful:

- create new array with 2x capacity.
- copy all old elements to new array.
- Delete the old array.
- Add the new element.

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Dymamic Array in Different language:

python - List.

Java - ArrayList.

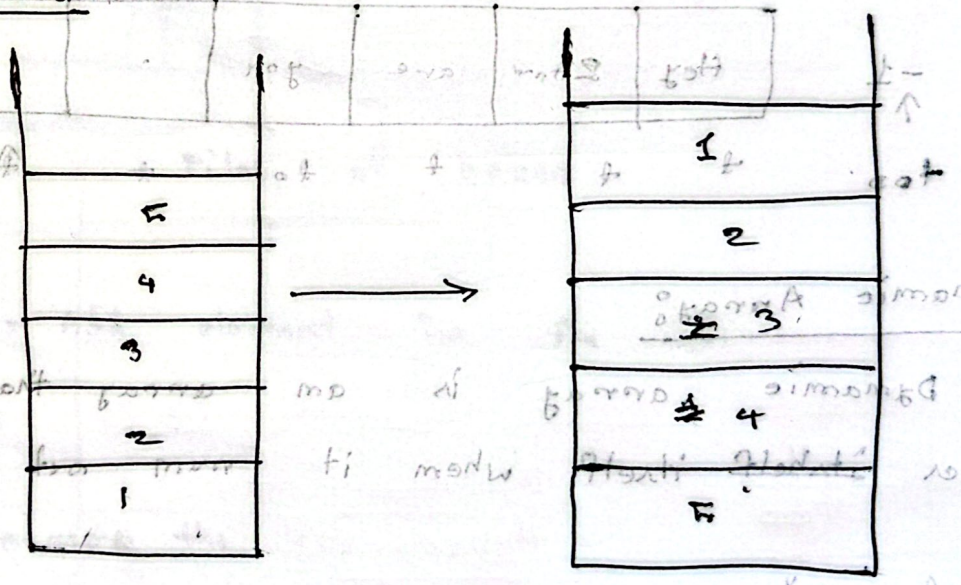
C++ = vector.

Reverse a Sentence using Stack:

Hey, How are you doing?

Doing, you are how hey?

Reverse a Stack:



Introduction to prefix, infix and postfix:

Infix Expression

operand — operator — operand

Precedence

1] ( ) { } [ ]

2] ^ — R → L

3] * , / L → R

4] + , − L → R

when the precedence of the operator is same.

Prefix Expression

#

operator — operand — operand

$((4 * 2) + 3) \longrightarrow + * 4 2 3$

$\cancel{(5-6)/3}$

$(5-(6/3)) \longrightarrow - 5/6 3$

Postfix Expression

operand — operand — operator

$(4 * 2) + 3$ ——— $4 2 * 3 +$

$5-(6/3)$ ——— $5 6 3 / -$

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